

Week03 – Jump Loop and Call + All Over IP
PC 12 รหัส B6615574 ชื่อ-สกุล นายธีระพัฒน์ แสงวงศ์
เปลี่ยนชื่อไฟล์เป็น “B6501234-นายไมโคร รายงานแลป-Lab03.pdf” ส่งงาน Report ที่ >> https://forms.gle/6cJ2JEeL3xp1pUCf8

3. Delay and Display on ETT-8052 Board

- 3.1 อ่านค่า oscillator frequency บนบอร์ด ET-8032 V2 = _____ MHz
- 3.2 คำนวณค่า oscillator period ของบอร์ด ET-8032 V2 = _____ μ Sec
- 3.3 คำนวณเวลาในรอบการทำงานของบอร์ด ET-8032 V2 หรือ 1MC = _____ μ Sec
- 3.4 โปรแกรม “Lab33.Asm” เป็นโปรแกรมสำหรับแสดงค่า 0 ถึง F ที่พอร์ต 7_Segment ตำแหน่ง 0E060H ทดสอบการทำงานด้วย ET-8032 V2 Board
- 3.5 ให้นักศึกษาปรับแก้โปรแกรมในข้อที่ 3.4 เพื่อให้การแสดงผลตัวอักษรที่ตัวแสดงผล 7 ส่วน (7-segment display) เป็นรหัสนักศึกษาเอง และใช้การหน่วงเวลา 300+รหัสนักศึกษาสองตัวท้าย หน่วยเป็น มิลลิวินาที ให้ใช้ delay ดังต่อไปนี้ และ **ให้แสดงวิธีการคำนวณก่อน □ ตรวจสอบ**

My_DELAY: PUSH 07H ; MC= PUSH 06H ; MC= PUSH 05H ; MC= PUSH 04H ; MC= MOV R4,#xx ; MC= _DLY_4: MOV R5,#ss ; MC= _DLY_5: MOV R6,#uu ; MC= _DLY_6: MOV R7,#tt ; MC= _DLY_7: DJNZ R7,_DLY_7 ; MC= DJNZ R6,_DLY_6 ; MC= DJNZ R5,_DLY_5 ; MC= DJNZ R4,_DLY_4 ; MC= POP 04H ; MC= POP 05H ; MC= POP 06H ; MC= POP 07H ; MC= RET ; MC	ก. หน่วงเวลา 300+รหัสนักศึกษาสองตัวท้าย หน่วยเป็น มิลลิวินาที ข. หน่วงเวลา(วินาที) = _____ ค. คิดเป็นจำนวน MC = _____ = _____ = _____ ง. จาก My_DELAY สร้างสมการเพื่อคำนวณ Machine Cycle เมื่อติดตัวแปร xx, ss, uu, tt เป็นอย่างไร และ คำนวณหาค่าตัวแปร xx, ss, uu, tt ในโปรแกรมมัลดูป
--	---

Your ESP32 MAC ID =

Loop	Loop + Condition	Delay 300+xx mS	Web 4 LED	Disp. + NTP + DHT22


```

        ORG 0000H
        MOV R7,#00H ; Start Read RAM
        MOV DPTR,#9000H; Save RAM

LOOP:   MOV A,R7
        MOVX @DPTR,A
        INC R7
        INC R7
        INC DPTR
        MOV A,DPH
        CJNE A,#92H,Loop ; Stop 7FH+1
        JMP $

        END

```

Text Code

```

        ORG 0000H
        MOV R7,#00H ; Start Read RAM
        MOV DPTR,#9000H; Save RAM

LOOP:   MOV A,R7
        MOVX @DPTR,A
        INC R7
        INC R7
        INC DPTR
        MOV A,DPH
        CJNE A,#92H,Loop ; Stop 7FH+1
        JMP $

        END

```

2. Loop + Condition and Data Transfer (TS Emulator)

- 2.1 ให้นักศึกษาเขียนโปรแกรมเพื่อตรวจสอบว่าค่าข้อมูลที่อยู่ในหน่วยความจำภายในตำแหน่ง 40H-7FH ว่ามีค่าเท่ากับค่าที่เก็บในรีจิสเตอร์ B หรือไม่โดยเก็บค่าที่นับได้ในหน่วยความจำภายนอกตำแหน่ง 9000H
- 2.2 ให้นักศึกษาเขียนโปรแกรมย้ายข้อมูลในหน่วยความจำ 30H ถึง 7FH ไปไว้ที่หน่วยความจำภายนอกตำแหน่งที่ 9000H เป็นต้นไป โดยกำหนดเงื่อนไขการย้าย คือ หากข้อมูลมีค่า เป็น 00 ปลายทางจะกำหนดให้เป็น 0FFH

Picture(รูปถ่ายการทำงาน) – ข้อ 2

The screenshot displays the TS Emulator 8051 Evaluation Version 1.00 - TS Controls interface. The main window is divided into several panes:

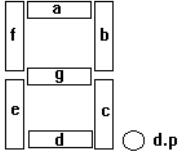
- Left Pane:** Shows the register status. B is 10, ACC is 05, DPTR is 9100, SP is 07 [00], PSW is 00 ---0---, PC is 0018, R0 is 20, R1 is 00, R2 is 00, R3 is 05, R4 is 00, and R5 is 00.
- Disassembled Code Pane:** Shows the assembly code. The current instruction is `0018 80 fe SJMP 0018`, which is highlighted. Other instructions include `000c a3 INC DPTR`, `000d b5 f0 01 CJNE A,B,0011`, `0010 0b INC R3`, `0011 e5 83 MOV A,DPH`, `0013 b4 91 f5 CJNE A,#91,000B`, `0016 eb MOV A,R3`, `0017 f6 MOV @R0,A`, `001a 00 NOP`, `001b 00 NOP`, `001c 00 NOP`, and `001d 00 NOP`.
- Internal RAM Pane:** Shows the memory contents. The address 0018 contains the value 00fe. The address 0019 contains the value 0000. The address 0020 contains the value 0000. The address 0030 contains the value 0000. The address 0040 contains the value 0000. The address 0050 contains the value 0000.
- External RAM Pane:** Shows the memory contents. The address 9000 contains the value 0000. The address 9010 contains the value 0000. The address 9020 contains the value 0000. The address 9030 contains the value 0000. The address 9040 contains the value 0000. The address 9050 contains the value 0000. The address 9060 contains the value 0000. The address 9070 contains the value 0000. The address 9080 contains the value 0000. The address 9090 contains the value 0000. The address 90a0 contains the value 0000. The address 90b0 contains the value 0000. The address 90c0 contains the value 0000. The address 90d0 contains the value 0000. The address 90e0 contains the value 0000. The address 90f0 contains the value 0000. The address 9100 contains the value 0000.
- Source Listing Pane:** Shows the source code. The address 0018 contains the value 00fe. The address 0019 contains the value 0000. The address 0020 contains the value 0000. The address 0030 contains the value 0000. The address 0040 contains the value 0000. The address 0050 contains the value 0000.

Capture Code

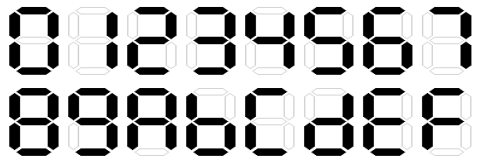
	ORG	0000H
	MOV	B, #10H
	MOV	R0, #20H
	MOVX	@DPTR, A
	MOV	R3, #00H
	MOV	DPTR, #9000H
LOOP:	MOVX	A, @DPTR
	INC	DPTR
	CJNE	A, B, LOOP1
	INC	R3
LOOP1:	MOV	A, DPH
	CJNE	A, #91H, LOOP
	MOV	A, R3
	MOV	@R0, A
	JMP	\$
	END	
Text Code		
	ORG	0000H
	MOV	B, #10H
	MOV	R0, #20H
	MOVX	@DPTR, A
	MOV	R3, #00H
	MOV	DPTR, #9000H
LOOP:	MOVX	A, @DPTR
	INC	DPTR
	CJNE	A, B, LOOP1
	INC	R3
LOOP1:	MOV	A, DPH
	CJNE	A, #91H, LOOP
	MOV	A, R3
	MOV	@R0, A
	JMP	\$
	END	

Arduino ESP32 7Segment Test

- ต่อ ESP32 กับ Single Digit 7Segment แล้วเขียนโปรแกรมให้นับเลขดังนี้ (Arduino)



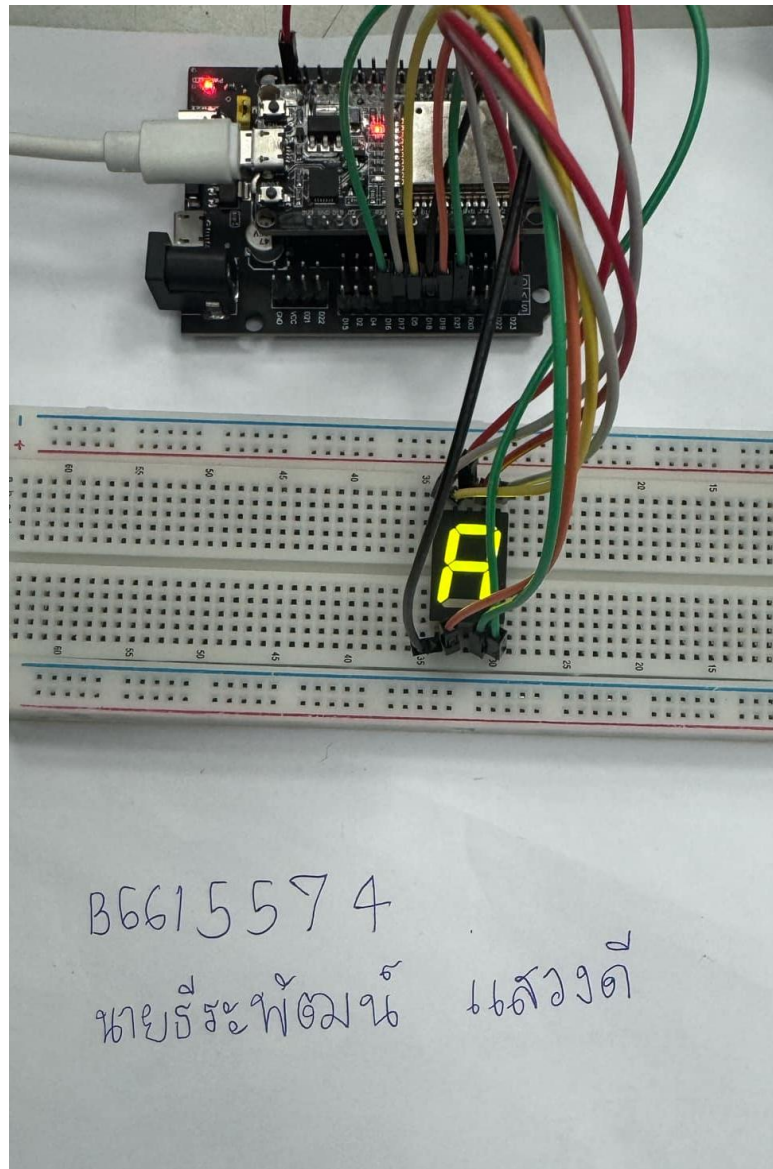
- ให้เรียงบิตเป็น tgfe dcba
- ทดสอบโปรแกรมให้นับลง F → 0
- แต่ละค่าห่างกัน 200 mSec
- ปรับแก้เป็น grade_digital=A



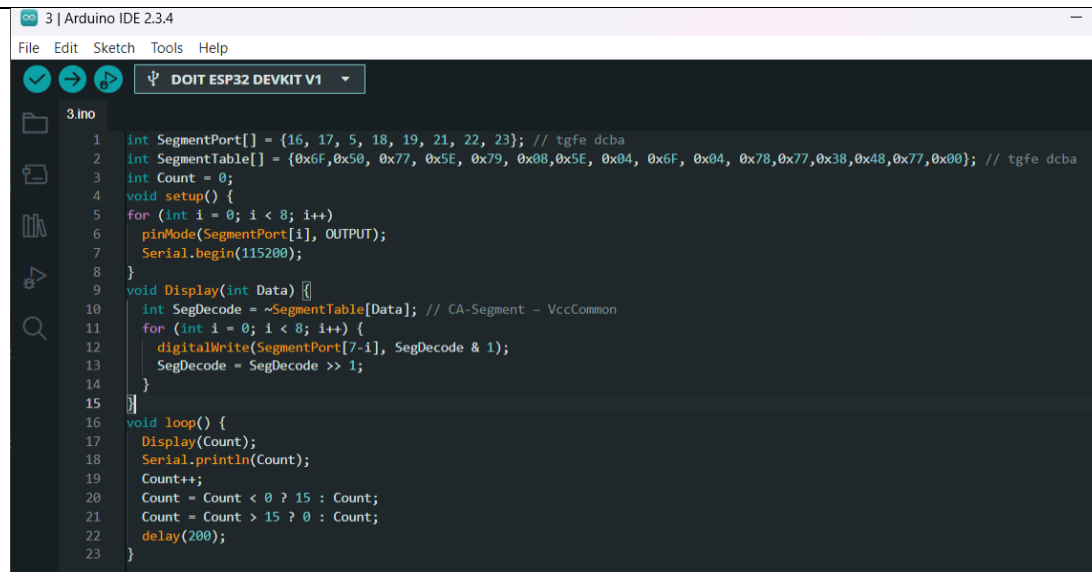
0000	0001	0010	0011	0100	0101	0110	0111
9	-	A	d	E	-	d	.

1000	1001	1010	1011	1100	1101	1110	1111
9	.	E	A	L	=	A	-

Picture(รูปถ่ายการทำงาน) – ข้อ Single Digit 7 Segment



Capture Code



```

3 | Arduino IDE 2.3.4
File Edit Sketch Tools Help

DOIT ESP32 DEVKIT V1

3.ino
1  int SegmentPort[] = {16, 17, 5, 18, 19, 21, 22, 23}; // tgfe dcba
2  int SegmentTable[] = {0x6F,0x50, 0x77, 0x5E, 0x79, 0x08,0x5E, 0x04, 0x6F, 0x04, 0x78,0x77,0x38,0x48,0x77,0x00}; // tgfe dcba
3  int Count = 0;
4  void setup() {
5      for (int i = 0; i < 8; i++)
6          pinMode(SegmentPort[i], OUTPUT);
7      Serial.begin(115200);
8  }
9  void Display(int Data) {
10     int SegDecode = ~SegmentTable[Data]; // CA-Segment - VccCommon
11     for (int i = 0; i < 8; i++) {
12         digitalWrite(SegmentPort[7-i], SegDecode & 1);
13         SegDecode = SegDecode >> 1;
14     }
15 }
16 void loop() {
17     Display(Count);
18     Serial.println(Count);
19     Count++;
20     Count = Count < 0 ? 15 : Count;
21     Count = Count > 15 ? 0 : Count;
22     delay(200);
23 }

```

Text Code

```

int SegmentPort[] = {16, 17, 5, 18, 19, 21, 22, 23}; // tgfe dcba
int SegmentTable[] = {0x6F,0x50, 0x77, 0x5E, 0x79, 0x08,0x5E, 0x04, 0x6F, 0x04, 0x78,0x77,0x38,0x48,0x77,0x00}; // tgfe dcba
int Count = 0;
void setup() {
    for (int i = 0; i < 8; i++)
        pinMode(SegmentPort[i], OUTPUT);
    Serial.begin(115200);
}
void Display(int Data) {
    int SegDecode = ~SegmentTable[Data]; // CA-Segment - VccCommon
    for (int i = 0; i < 8; i++) {
        digitalWrite(SegmentPort[7-i], SegDecode & 1);
        SegDecode = SegDecode >> 1;
    }
}
void loop() {
    Display(Count);
    Serial.println(Count);
    Count++;
    Count = Count < 0 ? 15 : Count;
    Count = Count > 15 ? 0 : Count;
    delay(200);
}

```


ตรวจสอบและบันทึก ESP32 MAC ID ของตัวเอง

Picture(รูปถ่ายการทำงาน) – ข้อ MAC Address

```

Output  Serial Monitor X
Message (Enter to send message to 'DOIT ESP32 DEVKIT V1' on 'COM4')

.....ESP32 Chip ID = 4C:11:AE:70:46:EC
ets Jun  8 2016 00:22:57

rst:0x1 (POWERON_RESET),boot:0x13 (SPI_FAST_FLASH_BOOT)
configsip: 0, SPIWP:0xee
clk_drv:0x00,q_drv:0x00,d_drv:0x00,cs0_drv:0x00,hd_drv:0x00,wp_drv:0x00
mode:DIO, clock div:1
load:0x3fff0030,len:4688
load:0x40078000,len:15516
load:0x40080400,len:4
load:0x40080404,len:3196
entry 0x400805a4
ESP32 Chip ID = 4C:11:AE:70:46:EC
ESP32 Chip ID = 4C:11:AE:70:46:EC
ESP32 Chip ID = 4C:11:AE:70:46:EC
ESP32 Chip ID = 4C:11:AE:70:46:EC
ESP32 Chip ID = 4C:11:AE:70:46:EC
ESP32 Chip ID = 4C:11:AE:70:46:EC

```

Capture Code

2_9 | Arduino IDE 2.3.4

File Edit Sketch Tools Help

DOIT ESP32 DEVKIT V1

2_9.ino

```

1  uint64_t chipid;
2  void setup() {
3    Serial.begin(115200);
4  }
5  void loop() {
6    chipid = ESP.getEfuseMac(); //The chip ID is essentially its MAC address(length: 6 bytes).
7    Serial.printf("ESP32 Chip ID = ");
8    Serial.printf("%02X:", (uint8_t)(chipid >> 0));
9    Serial.printf("%02X:", (uint8_t)(chipid >> 8));-
10   Serial.printf("%02X:", (uint8_t)(chipid >> 16));
11   Serial.printf("%02X:", (uint8_t)(chipid >> 24));
12   Serial.printf("%02X:", (uint8_t)(chipid >> 32));
13   Serial.printf("%02X", (uint8_t)(chipid >> 40));
14   Serial.println();
15   delay(3000);
16 }

```

Text Code

```

uint64_t chipid;
void setup() {
  Serial.begin(115200);
}
void loop() {
  chipid = ESP.getEfuseMac(); //The chip ID is essentially its MAC address(length: 6 bytes).
  Serial.printf("ESP32 Chip ID = ");
  Serial.printf("%02X:", (uint8_t)(chipid >> 0));
  Serial.printf("%02X:", (uint8_t)(chipid >> 8));-
  Serial.printf("%02X:", (uint8_t)(chipid >> 16));
  Serial.printf("%02X:", (uint8_t)(chipid >> 24));
  Serial.printf("%02X:", (uint8_t)(chipid >> 32));
  Serial.printf("%02X", (uint8_t)(chipid >> 40));
  Serial.println();
  delay(3000);
}

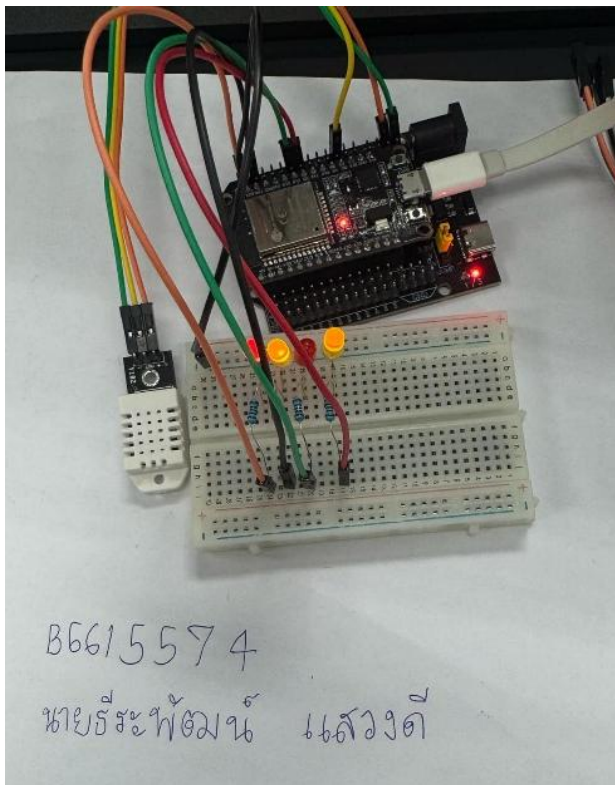
```

4. ทดสอบการใช้ ESP32 เป็น Web Server

- 4.1 อยากมีกด Link ไปที่หน้า FB ของตัวเอง
- 4.2 และปุ่มสำหรับคุมปิด-เปิด หลอดไฟ LED 4 ดวง



Picture(รูปถ่ายการทำงาน) – ข้อ 4



B6615574

นายธีระพัฒน์ แก้ววงศ์

Capture Code

```

4 - Test AJAX_01.ino | Arduino IDE 2.3.4
File Edit Sketch Tools Help

DOIT ESP32 DEVKIT V1

4.ino Test AJAX_01.ino index.h
1 #include <WiFi.h>
2 #include <WiFiClient.h>
3 #include <WebServer.h>
4 #include "DHTesp.h"
5 #include "index.h" //Our HTML webpage contents with javascripts
6 #define DHT_Pin 15
7 #define testLED1 21
8 #define testLED2 19
9 #define testLED3 18
10 #define testLED4 5
11 const char* ssid = "Toto";
12 const char* password = "toto55555";
13 WebServer server(80); //Server on port 80
14 DHTesp dht;
15 String ledState1 = "NA";
16 String ledState2 = "NA";
17 String ledState3 = "NA";
18 String ledState4 = "NA";
19 void handleRoot() {
20   String s = MAIN_page; //Read HTML contents
21   server.send(200, "text/html", s); //Send web page
22 }
23 void handle_DHT22() {
24   float h = dht.getHumidity();
25   float t = dht.getTemperature();
26   String tmpValue = "Temp = ";
27   tmpValue += String(t) + " C, Humidity = ";
28   tmpValue += String(h) + " %";
29   server.send(200, "text/plain", tmpValue); //Send value to client ajax request
30 }
31 void handleLED() {
32   String t_state = server.arg("LEDstate"); //Refer xhttp.open("GET", "setLED?LEDstate="+led, true);
33   Serial.println(t_state);
34   if (t_state == "11") { digitalWrite(testLED1, HIGH); ledState1 = "ON"; } //Feedback parameter
35   if (t_state == "10") { digitalWrite(testLED1, LOW); ledState1 = "OFF"; } //Feedback parameter
36   if (t_state == "21") { digitalWrite(testLED2, HIGH); ledState2 = "ON"; } //Feedback parameter
37   if (t_state == "20") { digitalWrite(testLED2, LOW); ledState2 = "OFF"; } //Feedback parameter
38   if (t_state == "31") { digitalWrite(testLED3, HIGH); ledState3 = "ON"; } //Feedback parameter
39   if (t_state == "30") { digitalWrite(testLED3, LOW); ledState3 = "OFF"; } //Feedback parameter
40   if (t_state == "41") { digitalWrite(testLED4, HIGH); ledState4 = "ON"; } //Feedback parameter
41   if (t_state == "40") { digitalWrite(testLED4, LOW); ledState4 = "OFF"; } //Feedback parameter
42   server.send(200, "text/plain", ledState1+", "+ledState2+", "+ledState3+", "+ledState4); //Send web page
43 }
44 void setup(void) {
45   Serial.begin(115200);
46   dht.setup(DHT_Pin, DHTesp::DHT22);
47   pinMode(testLED1, OUTPUT);
48   pinMode(testLED2, OUTPUT);
49   pinMode(testLED3, OUTPUT);
50   pinMode(testLED4, OUTPUT);
51   Serial.print("\n\nConnect to ");
52   Serial.println(ssid);
53   WiFi.begin(ssid, password);
54   while (WiFi.status() != WL_CONNECTED) {
55     delay(500); Serial.print(".");
56   }
57   Serial.print("\n\nConnected "); Serial.println(ssid);
58   Serial.print("IP address: "); Serial.println(WiFi.localIP());
59   server.on("/", handleRoot);
60   server.on("/setLED", handleLED);
61   server.on("/read_DHT22", handle_DHT22);
62   server.begin();
63   Serial.println("HTTP server started");
64 }
65 void loop(void) {
66   server.handleClient(); //Handle client requests
67 }

```

```

4 - index.h | Arduino IDE 2.3.4
File Edit Sketch Tools Help

DOIT ESP32 DEVKIT V1

4.ino Test AJAX_01.ino index.h
1 const char MAIN_page[] PROGMEM = R"=====(
2 <!DOCTYPE html>
3 <html>
4 <body>
5 <div id="demo"> |
6 <h1>The ESP-32 Update web page without refresh</h1>
7 <button type="button" onclick="sendData(11)" style="background: rgb(202, 60, 60);">LED1 ON</button>
8 <button type="button" onclick="sendData(21)" style="background: rgb(202, 60, 60);">LED2 ON</button>
9 <button type="button" onclick="sendData(31)" style="background: rgb(202, 60, 60);">LED3 ON</button>
10 <button type="button" onclick="sendData(41)" style="background: rgb(202, 60, 60);">LED4 ON</button> <br><br>
11 <button type="button" onclick="sendData(10)" style="background: rgb(100,116,255);">LED1 OFF</button>
12 <button type="button" onclick="sendData(20)" style="background: rgb(100,116,255);">LED2 OFF</button>
13 <button type="button" onclick="sendData(30)" style="background: rgb(100,116,255);">LED3 OFF</button>
14 <button type="button" onclick="sendData(40)" style="background: rgb(100,116,255);">LED4 OFF</button> <br><br>
15 State of [LED1, LED2, LED3, LED4] is >> <span id="LEDState">NA</span><br>
16 </div>
17 <div>
18 <br>DHT-22 sensor : <span id="DHT22_Value">0</span><br>
19 </div>
20 <script>
21 function sendData(led) {
22   var xhttp = new XMLHttpRequest();
23   xhttp.onreadystatechange = function() {
24     if (this.readyState == 4 && this.status == 200) {
25       document.getElementById("LEDState").innerHTML =
26         this.responseText;
27     }
28   };
29   xhttp.open("GET", "setLED?LEDState="+led, true);
30   xhttp.send();
31 }
32 setInterval(function() {
33   // Call a function repetatively with 2 Second interval
34   getData();
35 }, 2000); //2000mSeconds update rate
36 function getData() {
37   var xhttp = new XMLHttpRequest();
38   xhttp.onreadystatechange = function() {
39     if (this.readyState == 4 && this.status == 200) {
40       document.getElementById("DHT22_Value").innerHTML =
41         this.responseText;
42     }
43   };
44   xhttp.open("GET", "read_DHT22", true);
45   xhttp.send();
46 }
47 </script>
48 <br><a href="https://web.facebook.com/theerapat.sawangdee/">Theerapat Sawangdee </a>
49 </body>
50 </html>
51 )=====";

```

Text Code	
Code <pre> #include <WiFi.h> #include <WiFiClient.h> #include <WebServer.h> #include "DHTesp.h" #include "index.h" //Our HTML webpage contents with javascripts #define DHT_Pin 15 #define testLED1 21 #define testLED2 19 #define testLED3 18 #define testLED4 5 const char* ssid = "Toto"; const char* password = "toto55555"; WebServer server(80); //Server on port 80 DHTesp dht; String ledState1 = "NA"; String ledState2 = "NA"; String ledState3 = "NA"; String ledState4 = "NA"; void handleRoot() { String s = MAIN_page; //Read HTML contents server.send(200, "text/html", s); //Send web page } void handle_DHT22() { float h = dht.getHumidity(); float t = dht.getTemperature(); String tmpValue = "Temp = "; tmpValue += String(t) + " C, Humidity = "; tmpValue += String(h) + " %"; server.send(200, "text/plain", tmpValue); //Send value to client ajax request } void handleLED() { String t_state = server.arg("LEDstate"); //Refer xhttp.open("GET", "setLED?LEDstate="+led, true); Serial.println(t_state); if (t_state == "11") { digitalWrite(testLED1, HIGH); ledState1 = "ON"; } //Feedback parameter if (t_state == "10") { digitalWrite(testLED1, LOW); ledState1 = "OFF"; } //Feedback parameter if (t_state == "21") { digitalWrite(testLED2, HIGH); ledState2 = "ON"; } //Feedback parameter if (t_state == "20") { digitalWrite(testLED2, LOW); ledState2 = "OFF"; } //Feedback parameter if (t_state == "31") { digitalWrite(testLED3, HIGH); ledState3 = "ON"; } //Feedback parameter if (t_state == "30") { digitalWrite(testLED3, LOW); ledState3 = "OFF"; } //Feedback parameter if (t_state == "41") { digitalWrite(testLED4, HIGH); ledState4 = "ON"; } //Feedback parameter if (t_state == "40") { digitalWrite(testLED4, LOW); ledState4 = "OFF"; } //Feedback parameter server.send(200, "text/plain", ledState1+", "+ledState2+", "+ledState3+", "+ledState4); //Send web page } void setup(void) { Serial.begin(115200); dht.setup(DHT_Pin, DHTesp::DHT22); </pre>	Index.h <pre> const char MAIN_page[] PROGMEM = R"=====(<DOCTYPE html> <html> <body> <div id="demo"> <h1>The ESP-32 Update web page without refresh</h1> <button type="button" onclick="sendData(11)" style="background: rgb(202, 60, 60);">LED1 ON</button> <button type="button" onclick="sendData(21)" style="background: rgb(202, 60, 60);">LED2 ON</button> <button type="button" onclick="sendData(31)" style="background: rgb(202, 60, 60);">LED3 ON</button> <button type="button" onclick="sendData(41)" style="background: rgb(202, 60, 60);">LED4 ON</button>

 <button type="button" onclick="sendData(10)" style="background: rgb(100,116,255);">LED1 OFF</button> <button type="button" onclick="sendData(20)" style="background: rgb(100,116,255);">LED2 OFF</button> <button type="button" onclick="sendData(30)" style="background: rgb(100,116,255);">LED3 OFF</button> <button type="button" onclick="sendData(40)" style="background: rgb(100,116,255);">LED4 OFF</button>

 State of [LED1, LED2, LED3, LED4] is >> NA
 </div> <div>
DHT-22 sensor : 0
 </div> <script> function sendData(led) { var xhttp = new XMLHttpRequest(); xhttp.onreadystatechange = function() { if (this.readyState == 4 && this.status == 200) { document.getElementById("LEDState").innerHTML = this.responseText; } }; xhttp.open("GET", "setLED?LEDstate="+led, true); xhttp.send(); } setInterval(function() { // Call a function repetatively with 2 Second interval getData(); }, 2000); //2000mSeconds update rate function getData() { var xhttp = new XMLHttpRequest(); xhttp.onreadystatechange = function() { if (this.readyState == 4 && this.status == 200) { </pre>

<pre> pinMode(testLED1, OUTPUT); pinMode(testLED2, OUTPUT); pinMode(testLED3, OUTPUT); pinMode(testLED4, OUTPUT); Serial.print("\n\nConnect to "); Serial.println(ssid); WiFi.begin(ssid, password); while (WiFi.status() != WL_CONNECTED) { delay(500); Serial.print("."); } Serial.print("\nConnected "); Serial.println(ssid); Serial.print("IP address: "); Serial.println(WiFi.localIP()); server.on("/", handleRoot); server.on("/setLED", handleLED); server.on("/read_DHT22", handle_DHT22); server.begin(); Serial.println("HTTP server started"); } void loop(void) { server.handleClient(); //Handle client requests } </pre>	<pre> document.getElementById("DHT22_Value").innerHTML TML = this.responseText; } }; xhttp.open("GET", "read_DHT22", true); xhttp.send(); } </script>
Theerapat Sawangdee </body> </html>)====="; </pre>
---	---

5. ทดสอบการใช้ ESP32 เพื่ออ่านเวลาผ่าน NTP

5.1 อยากทำนาฬิกาแสดงผลที่ 7_Segment Display



- นักศึกษาชาย ใช้ MAX-7219 Display
- นักศึกษาหญิง ใช้ TM-1638 Display

Picture(รูปถ่ายการทำงาน) – ข้อ 5.1



Capture Code

```

5.1 | Arduino IDE 2.3.4
File Edit Sketch Tools Help

ψ DOIT ESP32 DEVKIT V1

5.1.ino
1  #include <NTPClient.h>
2  #include <WiFi.h>
3  #include <WiFiUdp.h>
4  #include "LedController.hpp"
5  #define PinDIN 19
6  #define PinCS 18
7  #define PinCLK 5
8  LedController<1,1> lc;
9  const char* ssid = "Toto";
10 const char* password = "toto55555";
11 WiFiUDP ntpUDP;
12 NTPClient timeClient(ntpUDP);
13 void setup() {
14     Serial.begin(115200);
15     WiFi.begin(ssid, password);
16     while ( WiFi.status() != WL_CONNECTED )
17     { delay ( 500 );
18       Serial.print ( "." );
19     }
20     timeClient.begin();
21     timeClient.setTimeOffset(25200); // +7Hour = (7H x 60Min x 60Sec = 25200 Sec)
22     Serial.println();
23     lc = LedController<1,1>(PinDIN, PinCLK, PinCS);
24     lc.setIntensity(8);
25     lc.clearMatrix();
26 }
27 void loop() {
28     timeClient.update();
29     Serial.print("Time -> "); Serial.print(timeClient.getEpochTime());
30     Serial.print(" -> "); Serial.println(timeClient.getFormattedTime());
31     int hour = timeClient.getHours(); // Get the hour
32     Serial.print(" -> Hour: "); Serial.println(hour);
33     int minute = timeClient.getMinutes(); // Get the minute
34     Serial.print(" -> Minute: "); Serial.println(minute);
35     int second = timeClient.getSeconds(); // Get the second
36     Serial.print(" -> Second: "); Serial.println(second);
37     lc.setDigit(0,7,hour/10,false);
38     lc.setDigit(0,6,hour%10,false);
39     lc.setChar(0,5,'-',false);
40     lc.setDigit(0,4,minute/10,false);
41     lc.setDigit(0,3,minute%10,false);
42     lc.setChar(0,2,'-',false);
43     lc.setDigit(0,1,second/10,false);
44     lc.setDigit(0,0,second%10,false);
45     delay(1000);
46 }
47

```


Text Code

```

#include <NTPClient.h>
#include <WiFi.h>
#include <WiFiUdp.h>
#include "LedController.hpp"
#define PinDIN 19
#define PinCS 18
#define PinCLK 5
LedController<1,1> lc;
const char* ssid = "Toto";
const char* password = "toto55555";
WiFiUDP ntpUDP;
NTPClient timeClient(ntpUDP);
void setup() {
  Serial.begin(115200);
  WiFi.begin(ssid, password);
  while ( WiFi.status() != WL_CONNECTED )
  { delay ( 500 );
    Serial.print ( "." );
  }
  timeClient.begin();
  timeClient.setTimeOffset(25200); // +7Hour = (7H x 60Min x 60Sec = 25200 Sec)
  Serial.println();
  lc = LedController<1,1>(PinDIN, PinCLK,PinCS);
  lc.setIntensity(8);
  lc.clearMatrix();
}
void loop() {
timeClient.update();
  Serial.print("Time-> "); Serial.print(timeClient.getEpochTime());
  Serial.print("-> "); Serial.println(timeClient.getFormattedTime());
  int hour = timeClient.getHours(); // Get the hour
  Serial.print("-> Hour: "); Serial.println(hour);
  int minute = timeClient.getMinutes(); // Get the minute
  Serial.print("-> Minute: "); Serial.println(minute);
  int second = timeClient.getSeconds(); // Get the second
  Serial.print("-> Second: "); Serial.println(second);
  lc.setDigit(0,7,hour/10,false);
  lc.setDigit(0,6,hour%10,false);
  lc.setChar(0,5,'-',false);
  lc.setDigit(0,4,minute/10,false);
  lc.setDigit(0,3,minute%10,false);
  lc.setChar(0,2,'-',false);
  lc.setDigit(0,1,second/10,false);
  lc.setDigit(0,0,second%10,false);
  delay(1000);
}

```

5.2 ปรับปรุงให้แสดงเวลา สลับกับอุณหภูมิ กำหนดวินาทีที่ 5, 6, 7, 8, 9, 10, 11 ให้แสดงอุณหภูมิ



- นักศึกษาชายรหัสคู่ ใช้ MAX-7219 Display
ให้แสดงเวลา สลับกับอุณหภูมิเซลเซียส สลับกับความชื้น
- นักศึกษาชายรหัสเดี่ยว ใช้ MAX-7219 Display
ให้แสดงเวลา สลับกับอุณหภูมิฟาเรนไฮต์ สลับกับความชื้น
- นักศึกษาหญิงรหัสคู่ ใช้ TM-1638 Display
ให้แสดงเวลา สลับกับอุณหภูมิเซลเซียส สลับกับความชื้น
- นักศึกษาหญิงรหัสเดี่ยว ใช้ TM-1638 Display
ให้แสดงเวลา สลับกับอุณหภูมิฟาเรนไฮต์ สลับกับความชื้น

Hint:

- อ่านเวลา
 - → หากวินาทีเป็น 5,6,7,8,9 แสดง Tempp : 12.34 C (or 12.34 F)
 - → หากวินาทีเป็น 35,36,37,38,39 แสดง Humid : 56.78 H
 - → นอกนั้นแสดง HH-MM-SS
- Read <http://www.whatimade.today/programming-an-8-digit-7-segment-display-the-easy-way-using-a-max7219/>

Picture(รูปถ่ายการทำงาน) – ข้อ 5.2



Capture Code

```

5.2 | Arduino IDE 2.3.4
File Edit Sketch Tools Help

ψ DOIT ESP32 DEVKIT V1

5.2.ino
1 #include <NTPClient.h>
2 #include <WiFi.h>
3 #include <WiFiUdp.h>
4 #include "DHTesp.h"
5 #define PinDHT22 15
6 DHTesp dht;
7 #include "LedController.hpp"
8 #define PinDIN 19
9 #define PinCS 18
10 #define PinCLK 5
11 LedController<1,1> lc;
12 const char* ssid = "Toto";
13 const char* password = "toto55555";
14 WiFiUDP ntpUDP;
15 NTPClient timeClient(ntpUDP);
16 void setup() {
17   Serial.begin(115200);
18   WiFi.begin(ssid, password);
19   while ( WiFi.status() != WL_CONNECTED )
20   { delay ( 500 );
21     Serial.print ( "." );
22   }
23   timeClient.begin();
24   timeClient.setTimeOffset(25200); // +7Hour = (7H x 60Min x 60Sec = 25200 Sec)
25   Serial.println();
26   lc = LedController<1,1>(PinDIN, PinCLK,PinCS);
27   lc.setIntensity(8);
28   lc.clearMatrix();
29   dht.setup(PinDHT22,DHTesp::DHT22);
30 }
31 void loop() {
32   int intData,singleData;
33   float Humid = dht.getHumidity();
34   float Temp = dht.getTemperature();
35   timeClient.update();
36   Serial.print("Time -> "); Serial.print(timeClient.getEpochTime());
37   Serial.print(" -> "); Serial.println(timeClient.getFormattedTime());
38   int hour = timeClient.getHours(); // Get the hour
39   Serial.print(" -> Hour: "); Serial.println(hour);
40   int minute = timeClient.getMinutes(); // Get the minute
41   Serial.print(" -> Minute: "); Serial.println(minute);
42   int second = timeClient.getSeconds(); // Get the second
43   Serial.print(" -> Second: "); Serial.println(second);
44   if(second>=5 & second <=11){
45     intData = (int)(Temp*100);
46     lc.clearMatrix();
47     lc.setChar(0,0,'c',false);
48     for(int i=1; i<5;i++){
49       singleData=intData%10 ; intData/=10;
50       if(i==3){
51         lc.setDigit(0,i,singleData,true);
52       }else{
53         lc.setDigit(0,i,singleData,false);
54       }
55     }
56   }else if(second>=35 & second <=39){
57     intData = (int)(Humid*100);
58     lc.clearMatrix();
59     lc.setChar(0,0,'h',false);
60     for(int i=1; i<5;i++){
61       singleData=intData%10 ; intData/=10;
62       if(i==3){
63         lc.setDigit(0,i,singleData,true);
64       }else{
65         lc.setDigit(0,i,singleData,false);
66       }
67     }
68   }else{
69     lc.setDigit(0,7,hour/10,false);
70     lc.setDigit(0,6,hour%10,false);
71     lc.setChar(0,5,'-',false);
72     lc.setDigit(0,4,minute/10,false);
73     lc.setDigit(0,3,minute%10,false);
74     lc.setChar(0,2,'-',false);
75     lc.setDigit(0,1,second/10,false);
76     lc.setDigit(0,0,second%10,false);
77   }
78   delay(1000);
79 }

```

Text Code

```

#include <NTPClient.h>
#include <WiFi.h>
#include <WiFiUdp.h>
#include "DHTesp.h"
#define PinDHT22 15
DHTesp dht;
#include "LedController.hpp"
#define PinDIN 19
#define PinCS 18
#define PinCLK 5
LedController<1,1> lc;
const char* ssid = "Toto";
const char* password = "toto55555";
WiFiUDP ntpUDP;
NTPClient timeClient(ntpUDP);
void setup() {
  Serial.begin(115200);
  WiFi.begin(ssid, password);
  while ( WiFi.status() != WL_CONNECTED )
  { delay ( 500 );
    Serial.print ( "." );
  }
  timeClient.begin();
  timeClient.setTimeOffset(25200); // +7Hour = (7H x 60Min x 60Sec = 25200 Sec)
  Serial.println();
  lc = LedController<1,1>(PinDIN, PinCLK, PinCS);
  lc.setIntensity(8);
  lc.clearMatrix();
  dht.setup(PinDHT22, DHTesp::DHT22);
}
void loop() {
  int intData, singleData;
  float Humid = dht.getHumidity();
  float Tempp = dht.getTemperature();
  timeClient.update();
  Serial.print("Time-> "); Serial.print(timeClient.getEpochTime());
  Serial.print("-> "); Serial.println(timeClient.getFormattedTime());
  int hour = timeClient.getHours(); // Get the hour
  Serial.print("-> Hour: "); Serial.println(hour);
  int minute = timeClient.getMinutes(); // Get the minute
  Serial.print("-> Minute: "); Serial.println(minute);
  int second = timeClient.getSeconds(); // Get the second
  Serial.print("-> Second: "); Serial.println(second);
  if(second >= 5 & second <= 11){
    intData = (int)(Tempp*100);
    lc.clearMatrix();
    lc.setChar(0,0,'c',false);
    for(int i=1; i<5;i++){
      singleData=intData%10 ; intData/=10;
      if(i==3){
        lc.setDigit(0,i,singleData,true);
      }else{
        lc.setDigit(0,i,singleData,false);
      }
    }
  }
  }else if(second >= 35 & second <= 39){
    intData = (int)(Humid*100);
    lc.clearMatrix();
    lc.setChar(0,0,'h',false);
    for(int i=1; i<5;i++){
      singleData=intData%10 ; intData/=10;

```

```
if(i==3){
    lc.setDigit(0,i,singleData,true);
}else{
    lc.setDigit(0,i,singleData,false);
}
}
}else{
    lc.setDigit(0,7,hour/10,false);
    lc.setDigit(0,6,hour%10,false);
    lc.setChar(0,5,'-',false);
    lc.setDigit(0,4,minute/10,false);
    lc.setDigit(0,3,minute%10,false);
    lc.setChar(0,2,'-',false);
    lc.setDigit(0,1,second/10,false);
    lc.setDigit(0,0,second%10,false);
}
delay(1000);
}
```